

# Question 1

Q2

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Ques 1: Schmidt Orthogonal Method

$$x_1 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Sol<sup>n</sup>:

$$\text{Let, } y_1 = x_1 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

We have formula

$$y_2 = x_2 - \frac{[x_2 y_1]}{[y_1 y_1]} y_1$$

$$= \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{i}{2} \\ 0 \end{bmatrix}$$

$$\therefore y_2 = \begin{bmatrix} \frac{1}{2} \\ -\frac{i}{2} \\ i \end{bmatrix}$$

Now,  $y_3 = ?$

$$y_3 = x_3 - \frac{[x_3 y_1]}{[y_1 y_1]} y_1 - \frac{[x_3 y_2]}{[y_2 y_2]} y_2$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - \frac{0}{2} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} - \frac{-i}{\frac{3}{2}} \begin{bmatrix} \frac{1}{2} \\ -\frac{i}{2} \\ i \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - \frac{-i}{\frac{3}{2}} \begin{bmatrix} \frac{1}{2} \\ -\frac{i}{2} \\ i \end{bmatrix}$$

$$i^2 = -1 \& i = \sqrt{-1}$$

$$[x_2 y_1] = y_1^H x_2$$

$$= [1-i \ 0] \cdot \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= 1+0+0$$

$$= 1$$

$$[y_1 y_1] = y_1^H y_1$$

$$= [1-i \ 0] \cdot \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= 1-i^2+0$$

$$= 1-(-1)$$

$$= 2$$

$$x_3 y_1 = y_1^H x_3$$

$$= [1-i \ 0] \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= 0$$

$$x_3 y_2 = y_2^H x_3$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{i}{2} & -i \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \cdot 0 + 0 - i$$

$$= -i$$

$$y_2 y_2 = y_2^H y_2$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{i}{2} & -i \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} \\ -\frac{i}{2} \\ i \end{bmatrix}$$

$$= \frac{1}{4} - \frac{i^2}{4} - i^2 = \frac{1}{4} + \frac{1}{4} - (-1)$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - \left(-\frac{2i}{3}\right) \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \frac{2i}{3} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} \frac{i}{3} \\ -\frac{1}{3}i^2 \\ \frac{2i^2}{3} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} \frac{i}{3} \\ -\frac{1}{3} = \frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ 1 - \frac{2}{3} = \frac{3-2}{3} = \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

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$$\frac{1+1+4}{4} = \frac{6}{4} = \frac{3}{2}$$

Prove that, orthogonal to each other.

For example if  $Y_1 \perp Y_2$

$$\begin{aligned} [Y_1, Y_2] &= Y_2^H \cdot Y_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2}i & -i \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ 0 \end{bmatrix} \\ &= \frac{1}{2} + \frac{1}{2}i^2 - 0 \\ &= \frac{1}{2} - \frac{1}{2} - 0 \\ &= 0 \end{aligned}$$

Now,

$$Z_1 = \frac{Y_1}{\|Y_1\|_2} = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \cdot \frac{1}{\sqrt{(Y_1 \cdot Y_1)}} \quad \text{pg-27}$$

$$Z_1 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}}$$

And,

$$Z_2 = \frac{Z_2}{\|Z_2\|_2} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix} \cdot \frac{1}{\sqrt{(Z_2 \cdot Z_2)}}$$

$$= \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix} \cdot \frac{1}{\sqrt{\frac{3}{2}}}$$

$$Z_2 = \sqrt{\frac{2}{3}} \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2}i \\ i \end{bmatrix}$$

$$\begin{aligned} & \sqrt{(Y_1 \cdot Y_1)} \\ &= \sqrt{1^2 + 1^2 + 0^2} \\ &= \sqrt{2} \\ & \sqrt{Y_2 \cdot Y_2} \\ &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}i\right)^2 + i^2} \\ &= \sqrt{\frac{1}{4} + \frac{1}{4} + 1} \\ &= \sqrt{\frac{1+1+4}{4}} \\ &= \sqrt{\frac{3}{2}} \end{aligned}$$

Further we can simplify

$$\frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{1}{2} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} = \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{1}{\sqrt{2} \cdot \sqrt{2}} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

Then,

$$Z_3 = \frac{Y_3}{\|Y_3\|_2} = \begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix} \cdot \frac{1}{\sqrt{(Y_3 \cdot Y_3)}} \\ = \begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix} \cdot \frac{1}{\sqrt{\frac{1}{3}}}$$

$$\begin{aligned} & \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} \\ &= \sqrt{\frac{1+1+1}{9}} \\ &= \sqrt{\frac{3}{9}} \\ &= \sqrt{\frac{1}{3}} \end{aligned}$$

$$= \sqrt{3} \begin{bmatrix} \frac{i}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

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$$= \sqrt{3} \cdot \frac{1}{3} \begin{bmatrix} i \\ 1 \\ 1 \end{bmatrix} = \sqrt{3} \cdot \frac{1}{\sqrt{3} \cdot \sqrt{3}} \begin{bmatrix} i \\ 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} i \\ 1 \\ 1 \end{bmatrix}$$

We can check the utilization

①

$$(z_1, z_1) = z_1^H z_1$$

$$= \frac{1}{\sqrt{2}} [1 \ -i \ 0] \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

$$= \left(\frac{1}{\sqrt{2}}\right)^2 \cdot (1 + (-i^2) + 0)$$

$$= \frac{1}{2} \cdot (1 - i^2) = \frac{1}{2} (1 - (-1)) = \frac{1}{2} (1 + 1) = \frac{1}{2} \times 2 = 1$$

②

$$(z_2, z_2) = z_2^H z_2$$

$$= \frac{1}{\sqrt{6}} [1 \ i \ -2i] \cdot \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -i \\ 2i \end{bmatrix}$$

$$= \left(\frac{1}{\sqrt{6}}\right)^2 \cdot (1 + (-i^2) + (-4i^2))$$

$$= \frac{1}{6} \cdot (1 + (-1) + (-4 \cdot -1))$$

$$= \frac{1}{6} \cdot (1 + 1 + 4) = \frac{1}{6} \times 6 = 1$$

## Question 2

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Question: 2

$$A = \begin{bmatrix} 2i & 0 & -3 \\ 5 & 3 & 2 \\ -1 & 2 & 2 \end{bmatrix}; \text{Determine } m_1, F, m_\infty, 1, \infty$$

Sol<sup>n</sup>:

$$\textcircled{1} \|A\|_{m_1} = \sum_{i=1}^3 \sum_{j=1}^3 |a_{ij}|$$

$$A = \begin{bmatrix} 2i & 0 & -3 \\ 5 & 3 & 2 \\ -1 & 2 & 2 \end{bmatrix}$$

$$= |2i| + |0| + |-3| + |5| + |3| + |2| + |-1| + |2| + |2|$$

$$= 20$$

$$\textcircled{2} \|A\|_F = \sqrt{\sum_{i=1}^3 \sum_{j=1}^3 |a_{ij}|^2}$$

$$= \sqrt{|2i|^2 + |0|^2 + |-3|^2 + |5|^2 + |3|^2 + |2|^2 + |-1|^2 + |2|^2 + |2|^2}$$

$$= \sqrt{4 + 0 + 9 + 25 + 9 + 4 + 1 + 4 + 4}$$

$$= \sqrt{60}$$

$$\textcircled{3} \|A\|_{m_\infty} = 3 \cdot \max_{i,j} |a_{ij}|$$

$$= 3 \cdot 5$$

$$= 15$$

$$A = \begin{bmatrix} 2i & 0 & -3 \\ 5 & 3 & 2 \\ -1 & 2 & 2 \end{bmatrix}$$



4

$$\|A\|_1 = \max_j \sum_{i=1}^3 |a_{ij}|$$

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1<sup>st</sup> Column  $\Rightarrow 2 + |2| + 5 + |-1| = 8$  ↓ max.

2<sup>nd</sup> "  $\Rightarrow 0 + |3| + |2| = 5$

3<sup>rd</sup> "  $\Rightarrow |-3| + |2| + |2| = 7$

$\therefore \|A\|_1 = 8$

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 5 & 3 & 2 \\ -1 & 2 & 2 \end{bmatrix}$$

C      C<sup>column</sup>

8    5    7

5

$$\|A\|_\infty = \max_i \sum_{j=1}^3 |a_{ij}|$$

1<sup>st</sup> row  $\Rightarrow |2| + 0 + 3 = 5$

2<sup>nd</sup> "  $\Rightarrow |5| + |3| + |2| = 10$

3<sup>rd</sup> "  $\Rightarrow |-1| + |2| + |2| = 5$

$\|A\|_\infty = 10$

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 5 & 3 & 2 \\ -1 & 2 & 2 \end{bmatrix}$$

row

5    10    5

### Question 3

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Ques:- 3

Example-1 : For any  $A = (a_{ij})_{m \times n} \in \mathbb{C}^{m \times n}$ , we define

$$\|A\| = \sqrt{mn} \max_{i,j} |a_{ij}|$$

Prove  $\|\cdot\|_G$  is a kind of matrix in  $\mathbb{C}^{m \times n}$  and it's compatible with vector 2 norm.

Proof: (1) When  $A=O$ ;  $\|A\|_G = \sqrt{mn} \max_{i,j} |a_{ij}| = 0$   
 When  $A \neq O$ ; There is less one element  $a_{ij} \neq 0$

$$** \|A\|_G = \sqrt{mn} \max_{i,j} |a_{ij}| > 0$$

$$\begin{aligned} (2) \text{ For any } \lambda \in \mathbb{C}, ** \|\lambda A\|_G &= \sqrt{mn} \max_{i,j} |\lambda \cdot a_{ij}| \\ &= \sqrt{mn} \max_{i,j} |\lambda| \cdot |a_{ij}| \\ &= |\lambda| \cdot (\sqrt{mn} \max_{i,j} |a_{ij}|) \\ &= |\lambda| \cdot \|A\|_G \end{aligned}$$

(3) For any  $A, B \in \mathbb{C}^{m \times n}$ ;

$$\begin{aligned} \|A+B\|_G &= \sqrt{mn} \max_{i,j} |a_{ij} + b_{ij}| \\ &= \sqrt{mn} \max_{i,j} |a_{ij}| + \sqrt{mn} \max_{i,j} |b_{ij}| \\ &= \|A\|_G + \|B\|_G \end{aligned}$$

(4) For any  $A \in \mathbb{C}^{m \times n}$ ,  $B \in \mathbb{C}^{n \times l}$  [Pg-22]

$$\begin{aligned}
 \|AB\|_G &= \sqrt{ml} \max_{i,j} \left| \sum_{k=1}^n a_{ik} \cdot b_{kj} \right| \leq \sqrt{ml} \max_{i,j} \sum_{k=1}^n |a_{ik}| \cdot |b_{kj}| \\
 &\leq \sqrt{ml} \sum_{k=1}^n \left( \max_{ik} |a_{ik}| \cdot \max_{kj} |b_{kj}| \right) \\
 &\leq \sqrt{ml} \cdot n \max_{ik} |a_{ik}| \cdot \max_{kj} |b_{kj}| \\
 &= \left( \sqrt{mn} \cdot \max_{ik} |a_{ik}| \right) \cdot \left( \sqrt{nl} \max_{kj} |b_{kj}| \right) \\
 &= \|A\|_G \cdot \|B\|_G
 \end{aligned}$$

2nd part: (5) Now, we prove it is compatibility with vector 2-norm

For any  $x = (x_1, x_2, \dots, x_n)^T \neq 0$

$$\begin{aligned}
 \|Ax\|_2 &= \sqrt{\sum_{i=1}^m \left| \sum_{j=1}^n a_{ij} \cdot x_j \right|^2} \leq \sqrt{\sum_{i=1}^m \left( \sum_{j=1}^n |a_{ij}|^2 \right) \cdot \left( \sum_{j=1}^n |x_j|^2 \right)} \\
 &= \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2} \cdot \sqrt{\sum_{j=1}^n |x_j|^2} \\
 &\leq \sqrt{\sum_{i=1}^m \sum_{j=1}^n \max_{i,j} |a_{ij}|^2} \cdot \|x\|_2 = \sqrt{mn} \max_{i,j} |a_{ij}| \cdot \|x\|_2 \\
 &= \|A\|_G \cdot \|x\|_G
 \end{aligned}$$

Synthesize, the last proof (1)~(5).  
Hence,  $\|\cdot\|_G$  is a kind of matrix norm in  $\mathbb{C}^{m \times n}$ . And, it is compatible with vector 2-norm.



#### Question 4

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Ques-4 The condition number of a matrix

$$A = \begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix} = \begin{bmatrix} \swarrow & - \\ - & \searrow \end{bmatrix}$$

Calculate  $\text{cond}_1(A)$ ,  $\text{cond}_\infty(A)$ ,  $\text{cond}_2(A)$

$$\boxed{1} \text{ cond}_1(A) = \|A\|_1 \cdot \|A^{-1}\|_1$$

$$A^{-1} = \frac{\text{adj} A}{\det A} = \frac{\begin{bmatrix} 9 & -5 \\ -5 & 7 \end{bmatrix}}{63 - 25} = \frac{\begin{bmatrix} 9 & -5 \\ -5 & 7 \end{bmatrix}}{38} = \begin{bmatrix} \frac{9}{38} & -\frac{5}{38} \\ -\frac{5}{38} & \frac{7}{38} \end{bmatrix}$$

Compute

$$\|A\|_1$$

$$\text{Column 1 sum} = |7| + |5| = 12$$

$$\text{" 2 "} = |5| + |9| = 14$$

$$\text{So, } \|A\|_1 = 14$$

Compute

$$\|A^{-1}\|_1$$

$$\text{Column 1 sum} = \left| \frac{9}{38} \right| + \left| -\frac{5}{38} \right| = \frac{9+5}{38} = \frac{14}{38}$$

$$\text{Column 2 " } = \left| -\frac{5}{38} \right| + \left| \frac{7}{38} \right| = \frac{12}{38}$$

$$\text{So, } \|A^{-1}\|_1 = \frac{14}{38} = 0.3684$$

$$\text{Now, } \text{cond}_1(A) = \|A\|_1 \cdot \|A^{-1}\|_1$$

$$= 14 \times 0.3684 = 5.1579$$

2

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$$\text{cond}_\infty(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty$$

$$A^{-1} = \begin{bmatrix} \frac{9}{38} & -\frac{5}{38} \\ -\frac{5}{38} & \frac{7}{38} \end{bmatrix}$$

Compute

$$\|A\|_\infty = \max \left( \sum_{j=1}^n |a_{ij}| \right)$$

$$\text{Row 1 sum} = 12$$

$$\text{Row 2 " " } = 14$$

$$\text{So, } \|A\|_\infty = 14$$

Compute  $\|A^{-1}\|_\infty$

$$\text{Row 1 sum} = \frac{14}{38}$$

$$\text{" 2 " " } = \frac{12}{38}$$

$$\text{So, } \|A^{-1}\|_\infty = \frac{14}{38} = 0.3684$$

$\text{cond}_2(A)$  if  $A^T = A$  &  $\lambda_1 > \lambda_2 > 0$

$$\text{cond}_2(A) = \frac{\lambda_1}{\lambda_2}$$

$$A = \begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix}; A^T = A$$

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So,

Compute  $|\lambda I - A| = 0$

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}; \lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 7 & 5 \\ 5 & 9 \end{bmatrix}$$

Now,

$$|\lambda I - A| = \begin{vmatrix} \lambda - 7 & -5 \\ -5 & \lambda - 9 \end{vmatrix}$$

$$= (\lambda - 9)(\lambda - 7) - (-5)(-5)$$

$$= \lambda^2 - 7\lambda - 9\lambda + 63 - 25$$

$$= \lambda^2 - 16\lambda + 38$$

$$a = 1; b = -16; c = 38$$

Using quadratic formula

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(1)(38)}}{2 \cdot 1}$$

$$= \frac{16 \pm \sqrt{104}}{2} = \frac{16 \pm \sqrt{4 \times 26}}{2} = \frac{16 \pm 2\sqrt{26}}{2}$$

$$= 8 \pm \sqrt{26}$$

$$\lambda_1 = 8 + \sqrt{26} \quad \& \quad \lambda_2 = 8 - \sqrt{26}$$

$$\text{cond}_2(A) = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{8 + \sqrt{26}}{8 - \sqrt{26}}$$

### Question 5

Ques-5

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$$A = \begin{bmatrix} 3 & 1 & -1 \\ -2 & 0 & 2 \\ -1 & -1 & 3 \end{bmatrix}, \text{ calculate } e^{At}, \cos A$$

Sol<sup>n</sup>:

$$f(x) = |\lambda I - A|$$

$$= \begin{vmatrix} \lambda-3 & -1 & 1 \\ 2 & \lambda & -2 \\ 1 & 1 & \lambda-3 \end{vmatrix}$$

$$= (\lambda-3) [\lambda(\lambda-3) - (-2)] + 1 [2(\lambda-3) - (-2)] + 1 [2 - \lambda]$$

$$= (\lambda-3) (\lambda^2 - 3\lambda + 2) + (2\lambda - 6 + 2) + (2 - \lambda)$$

$$= \lambda^3 - 3\lambda^2 + 2\lambda - 3\lambda^2 + 9\lambda - 6 + 2\lambda - 4 + 2 - \lambda$$

$$= \lambda^3 - 6\lambda^2 + 12\lambda - 8$$

$$= \lambda^3 - 6\lambda^2 + 12\lambda - 2^3$$

$$= \lambda^3 - 3\lambda \cdot 2 + 3\lambda \cdot 2^2 - 2^3 \quad \left[ \because (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \right]$$

$$= (\lambda-2)^3 \text{ is polynomial characteristic}$$

Candidate for minimal polynomial

$$m(\lambda) = (\lambda-2), (\lambda-2)^2, (\lambda-2)^3$$

Eigenvalue in eq<sup>n</sup>

$$|\lambda I - A| = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_1 + R_3 \\ R_2 \rightarrow 2R_1 + R_2 \end{array}$$

$$= \begin{bmatrix} -1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



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Two eigenvector so 2 Jordan block

Pg-40

$$J = \begin{bmatrix} \boxed{2} & 1 & 0 \\ 0 & \boxed{2} & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} \boxed{2} & 0 & 0 \\ 0 & \boxed{2} & 1 \\ 0 & 0 & \boxed{2} \end{bmatrix}$$

$$m(\lambda) = (\lambda - 2)^2 \text{ minimal}$$

$$\text{Then, } r(\lambda) = b_1 \lambda + b_0$$

We calculate,  $e^{At}$ . So,  $f(At) = e^{At}$

$$\Rightarrow \Rightarrow \boxed{f(\lambda t) = e^{\lambda t}}$$

$$\boxed{\frac{d}{d\lambda} f(\lambda t) = t e^{\lambda t}}$$

$$\frac{d}{dt} f(\lambda t) = \frac{d}{d\lambda} e^{\lambda t} = t e^{\lambda t}$$

$$\text{Then, } \begin{cases} r(2) = 2b_1 + b_0 = f(2t) = e^{2t} \Rightarrow 2b_1 + b_0 = e^{2t} \\ \Rightarrow b_0 = e^{2t} - 2b_1 \text{ --- (i)} \end{cases}$$

$$\begin{cases} r'(2) = \frac{d}{dt} e^{2t} = e^{2t} \frac{d}{dt} (2t) = 2e^{2t} = b_1 \\ \therefore b_1 = te^{2t} \text{ --- (ii)} \end{cases}$$

$$\begin{cases} \text{(ii) in (i)} \\ b_0 = e^{2t} - 2 \cdot te^{2t} \\ \therefore b_0 = e^{2t} (1 - 2t) \text{ --- (iv)} \\ b_1 = te^{2t} \text{ --- (ii)} \end{cases}$$

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$$\begin{aligned}
 e^{At} &= b_1 A + b_0 I = t e^{2t} \begin{bmatrix} 3 & 1 & -1 \\ -2 & 0 & 2 \\ -1 & -1 & 3 \end{bmatrix} + (1-2t) e^{2t} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= e^{2t} \left( \begin{bmatrix} 3t & t & -t \\ -2t & 0 & 2t \\ -t & -t & 3t \end{bmatrix} + \begin{bmatrix} 1-2t & 0 & 0 \\ 0 & 1-2t & 0 \\ 0 & 0 & 1-2t \end{bmatrix} \right) \\
 &= e^{2t} \begin{pmatrix} 1+t & t & -t \\ -2t & 1-2t & 2t \\ -t & -t & 1+t \end{pmatrix}
 \end{aligned}$$

Calculate  $e^{\cos A}$ ,  $f(xt) = \cos(xt)$

$$\begin{aligned}
 \frac{d}{dx} f(xt) &= \frac{d}{dx} \cos(xt) \\
 &= -\sin(xt) \frac{d}{dx}(xt) \\
 \frac{d}{dx} f(xt) &= -\sin(xt) \cdot t
 \end{aligned}$$

Select,  $t=1$

$$\begin{aligned}
 \frac{d}{dx} f(x) &= -\sin(x) \\
 \text{Then, } \begin{cases} r(2) = 2b_1 + b_0 = f(2t) = \cos 2 \Rightarrow 2b_1 + b_0 = \cos 2 \\ r'(2) = -\sin(2) = b_1 \text{ --- (ii)} \Rightarrow b_0 = \cos 2 - 2b_1 \text{ --- (i)} \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 \text{Sol}^n: \quad & \begin{cases} b_0 = \cos 2 - 2(-\sin 2) \\ \quad = \cos 2 + 2\sin 2 \text{ --- (iv)} \\ \therefore b_0 = \cos 2 + 2\sin 2 \\ b_1 = -\sin 2 \text{ --- (iii)} \end{cases}
 \end{aligned}$$

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$$\cos A = b_1 A + b_0 I = -\sin 2A + (\cos 2 + 2\sin 2) I$$

$$= -\sin 2 \begin{bmatrix} 3 & 1 & -1 \\ -2 & 0 & 2 \\ -1 & -1 & 3 \end{bmatrix} + (\cos 2 + 2\sin 2) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -3\sin 2 & -\sin 2 & \sin 2 \\ 2\sin 2 & 0 & -2\sin 2 \\ \sin 2 & \sin 2 & -3\sin 2 \end{bmatrix} + \begin{bmatrix} \cos 2 + 2\sin 2 & 0 & 0 \\ 0 & \cos 2 + 2\sin 2 & 0 \\ 0 & 0 & \cos 2 + 2\sin 2 \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2 - \sin 2 & -\sin 2 & \sin 2 \\ 2\sin 2 & \cos 2 + 2\sin 2 & -2\sin 2 \\ \sin 2 & \sin 2 & \cos 2 - \sin 2 \end{bmatrix}$$

# Question 6

Example-6

Quest-6

Pg-15

$$e^{At} = \begin{bmatrix} 2e^{2t} - e^t & e^{2t} - e^t & e^t - e^{2t} \\ e^{2t} - e^t & 2e^{2t} - e^t & e^t - e^{2t} \\ 3e^{2t} - 3e^t & 3e^{2t} - 3e^t & 3e^t - 2e^{2t} \end{bmatrix}$$

We know,  $\frac{d}{dx} x^n = nx^{n-1}$

$$\text{So, } \frac{d}{dt}(2e^{2t} - e^t)$$

$$= \frac{d}{dt}(2e^{2t}) - \frac{d}{dt}(e^t)$$

$$= 2 \cdot \frac{d}{dt}(e^{2t}) - e^t$$

$$= 2 \cdot e^{2t} \cdot \frac{d}{dt}(2t) - e^t$$

$$= 2e^{2t} \cdot 2 \cdot \frac{d}{dt}(t) - e^t$$

$$2e^{2t} - e^t = 4e^{2t} - e^t$$

$$\frac{d}{dt}(e^{2t})$$

$$= e^{2t} \frac{d}{dt}(2t)$$

$$= e^{2t} \cdot 2 \cdot \frac{d}{dt}(t)$$

$$e^{2t} = e^{2t} \cdot 2$$

$$e^{2t} = 2e^{2t}$$

$$\frac{d}{dt}(3e^{2t})$$

$$3e^{2t} = 3 \frac{d}{dt}(e^{2t})$$

$$= 3e^{2t} \frac{d}{dt}(2t)$$

$$= 3e^{2t} \cdot 2 \cdot \frac{d}{dt}(t)$$

$$3e^{2t} = 6e^{2t}$$

$$\frac{d}{dt}(3e^t)$$

$$3e^t = 3e^t$$

$$\begin{bmatrix} 4e^{2t} - e^t & 2e^{2t} - e^t & e^t - 2e^{2t} \\ 2e^{2t} - e^t & 4e^{2t} - e^t & e^t - 2e^{2t} \\ 6e^{2t} - 3e^t & 6e^{2t} - 3e^t & 3e^t - 4e^{2t} \end{bmatrix}$$

$$t=0 \begin{bmatrix} 4e^0 - e^0 & 2e^0 - e^0 & e^0 - 2e^0 \\ 2e^0 - e^0 & 4e^0 - e^0 & e^0 - 2e^0 \\ 6e^0 - 3e^0 & 6e^0 - 3e^0 & 3e^0 - 4e^0 \end{bmatrix} = \begin{bmatrix} 4-1 & 2-1 & 1-2 \\ 2-1 & 4-1 & 1-2 \\ 6-3 & 6-3 & 3-4 \end{bmatrix}$$

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$$\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{bmatrix}$$



## Question 7

Ques-7

Pg-34

Rank

$$(1) \min(m, n)$$

$\downarrow$     $\downarrow$   
 $r$     $c$

(2) Check linear independent rows

Ques.: Find SVD

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}_{3 \times 2} \quad \min(3, 2) = 2$$

Solution: There are 3 steps: Use online calculator

~~Ques~~ (1)  $A^H = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  Step-1:  $A^H A$

So,

$$A^H A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1.1 + 0.0 + 1.1 & 1.0 + 0.1 + 1.1 \\ 0.1 + 1.0 + 1.1 & 0.0 + 1.1 + 1.1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Step(2): Find value of ' $\lambda$ '

$$|\lambda I - A| = ?$$

$$\begin{vmatrix} \lambda - 2 & -1 \\ -1 & \lambda - 2 \end{vmatrix}$$

$$= (\lambda - 2)(\lambda - 2) - (-1)(-1)$$

$$= (\lambda - 2)^2 - 1$$

$$= (\lambda^2 - 2 \cdot 2 \cdot \lambda + 2^2) - 1$$

$$= \lambda^2 - 4\lambda + 4 - 1$$

$$= \lambda^2 - 4\lambda + 3$$

$$= \lambda^2 - 3\lambda - \lambda + 3$$

$$= \lambda(\lambda - 3) - 1(\lambda - 3)$$

$$= (\lambda - 3)(\lambda - 1)$$

$$\therefore \lambda_1 = 3 \text{ \& } \lambda_2 = 1$$

Let,  $\lambda = 3$

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$$\text{So, } |3I - A| = \begin{vmatrix} 3-2 & -1 \\ -1 & 3-2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 \\ 0 & 0 \end{vmatrix} \quad R_2 \Rightarrow R_2 + R_1$$

$$\Rightarrow X_1 - X_2 = 0$$

Then,  $X_2 = 1$  then  $X_1 = X_2$   
then  $X_1 - 1 = 0$

$$\therefore X_1 = 1$$

$$P_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

And,  $\lambda = 1$

$$\text{Now, } |I - A| = \begin{vmatrix} 1-2 & -1 \\ -1 & 1-2 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -1 \\ -1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -1 \\ 0 & 0 \end{vmatrix} \quad R_1 = R_1 + (-R_2)$$

$$\Rightarrow -X_1 - X_2 = 0$$

$$\Rightarrow X_1 + X_2 = 0$$

$$\therefore X_1 = -X_2$$

And,  $X_2 = X_2$

$$P_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$V = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

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$$\text{Now, } \Sigma^r = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{bmatrix} = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{1} \end{bmatrix} = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore \Sigma^r^{-1} = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{bmatrix}$$

Step-3

$$U = AV \Sigma^r^{-1}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} + 0 & -\frac{1}{\sqrt{2}} + 0 \\ 0 + \frac{1}{\sqrt{2}} & 0 + \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \sqrt{2} & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}} + 0 & 0 + (-\frac{1}{\sqrt{2}}) \\ \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}} + 0 & 0 + \frac{1}{\sqrt{2}} \\ \frac{\sqrt{2}}{\sqrt{3}} + 0 & 0 + 0 \end{pmatrix}$$

$$= \begin{pmatrix} u_1 & u_2 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{\sqrt{2}}{\sqrt{3}} = \frac{2}{\sqrt{6}} & 0 \end{pmatrix} \quad \boxed{\text{pg-37}} \quad \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{2} \cdot \sqrt{2}}{\sqrt{3} \cdot \sqrt{2}} = \frac{2}{\sqrt{6}}$$

$$\frac{1}{\sqrt{6}} (1 \cdot x_1 + 1 \cdot x_2 + 2 \cdot x_3) = 0$$

$$\Rightarrow x_1 + x_2 + 2x_3 = 0$$

$$\frac{1}{\sqrt{2}} (-1 \cdot x_1 + 1 \cdot x_2) = 0$$

$$\Rightarrow -x_1 + x_2 = 0$$

$$\begin{pmatrix} 1 & 1 & 2 \\ -1 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{pmatrix} 0 & 2 & 2 \\ -1 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \rightarrow \frac{R_1}{2}} \begin{pmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} x_2 + x_3 = 0 \\ -x_1 + x_2 = 0 \end{cases} \Rightarrow \begin{cases} x_2 = -x_3 \\ x_1 = x_2 \\ \boxed{x_3 = x_3} \end{cases} \Rightarrow \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \Rightarrow u_3 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

Taking  $x_3 = 1$

$$U = \begin{pmatrix} \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^H$$

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$$A = \begin{pmatrix} \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$



# Question 8

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Ques-8

Let,  $A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Step-1

$$\psi(\lambda) = |\lambda I - A| = 0$$

$$\lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} \lambda-1 & -3 & -3 \\ -3 & \lambda-1 & -3 \\ -3 & -3 & \lambda-1 \end{bmatrix}$$

Now,

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\begin{bmatrix} \lambda-7 & \lambda-7 & \lambda-7 \\ -3 & \lambda-1 & -3 \\ -3 & -3 & \lambda-1 \end{bmatrix}$$

$$= (\lambda-7) \begin{bmatrix} 1 & 1 & 1 \\ -3 & \lambda-1 & -3 \\ -3 & -3 & \lambda-1 \end{bmatrix}$$

$$= (\lambda-7) \begin{bmatrix} 1 & 1 & 1 \\ 0 & \lambda+2 & 0 \\ 0 & 0 & \lambda+2 \end{bmatrix} \quad R_2 \rightarrow 3R_1 + R_2$$

$$= (\lambda-7) (\lambda+2)^2$$

$$\lambda_1 = 7, \lambda_2 = -2, -2$$

$$\lambda = (7, -2, -2)$$

$$\begin{aligned} 3 + \lambda - 1 \\ = \lambda + 2 \end{aligned}$$

For B

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$$\Psi(\lambda) = |\mu I - B|$$

$$\mu I = \begin{vmatrix} \mu & 0 & 0 & 0 \\ 0 & \mu & 0 & 0 \\ 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & \mu \end{vmatrix} \Rightarrow \mu I - B = \begin{vmatrix} \mu-1 & -1 & -1 & -1 \\ -1 & \mu-1 & -1 & -1 \\ -1 & -1 & \mu-1 & -1 \\ -1 & -1 & -1 & \mu-1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3 + R_4$$

$$|\mu I - B| = \begin{vmatrix} \mu-4 & \mu-4 & \mu-4 & \mu-4 \\ -1 & \mu-1 & -1 & -1 \\ -1 & -1 & \mu-1 & -1 \\ -1 & -1 & -1 & \mu-1 \end{vmatrix}$$

$$\boxed{R_2 \rightarrow R_2 + R_1} = (\mu-4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ -1 & \mu-1 & -1 & -1 \\ -1 & -1 & \mu-1 & -1 \\ -1 & -1 & -1 & \mu-1 \end{vmatrix}$$

$$R_2 \rightarrow R_2 + R_1; R_3 \rightarrow R_3 + R_1; R_4 \rightarrow R_4 + R_1$$

$$= (\mu-4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & \mu & 0 & 0 \\ 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & \mu \end{vmatrix}$$

$$= (\mu-4) \cdot \mu^3 \text{ So, } \mu=4, \mu=0,0,0$$

$$\text{So, } \mu = (4, 0, 0, 0)$$

$$\text{Now, } A \otimes B = \lambda \mu$$

$$\lambda = (7, -2, -2) \text{ \& } \mu = (4, 0, 0, 0)$$

$$\therefore \lambda \mu = (28, 0, 0, 0; -8, 0, 0, 0; -8, 0, 0, 0)$$